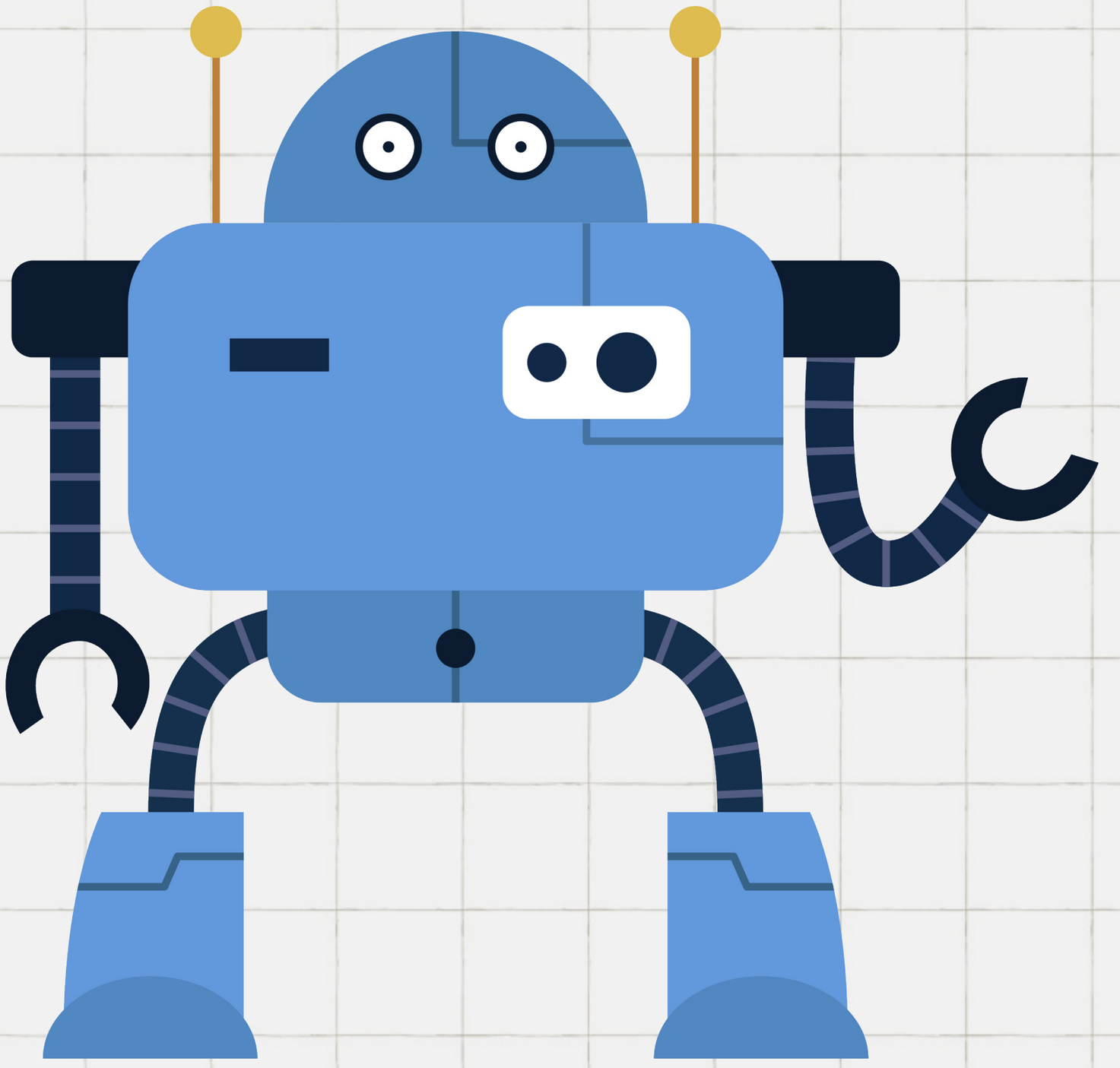
How the random module in python actually works



Computers are deterministic.

If you give a computer the same input and the same instructions, it must give you the same output.
So how does it act "random"?

It uses a

Pseudo random Number Generator (PRNG)

Pseudo means "fake"

It's a mathematical formula that produces a sequence of numbers that look random but are actually calculated.

A PRNG :

Starts with a seed (initial number).

Applies mathematical formula -> get current number

Repeat

That new number becomes the current number

So it looks like this :

Seed -> Number1 -> Number 2 -> Number 3 ...

Its basically a self feeding loop



The Seed : Initial value fed into a mathematical formula.

No Seed ? If you don't provide one , Python usually grabs the current system time , down to the microseconds, as the seed. Since time never stands still , every time you run your script, "seed" will be different

Mathematical formula may look something like this :

```
next_numb = (a × current_num
+ c) % m
```

This next number generated is produced as the "output"



But Python uses algorithm called The Mersenne Twister named after "Mersenne Primes"

The Seed (The Beginning): It starts with the Seed . This seed triggers a chain reaction to fill 624 empty boxes with initial numbers.

The Twist (The Mixing): When you ask for a number, it doesn't just pick a box. It "twists" bits from different boxes together using bitwise operations ensuring the pattern won't repeat for trillions of years.

The Result: It applies a final filter to the mix, pouring out a number that is perfectly balanced and unpredictable.

So if it´s all a Lie ? Can i build one myself .

Here s how i built a simple random generator of my own